

Day – 01

1. Basic Java program.

```
class First {  
    public static void main(String[] args) {  
        if (args.length > 0) {  
            System.out.println(args[0]);  
        } else {  
            System.out.println("No arguments provided");  
        }  
    }  
}
```

→ Javac Basic.java

→ Java Basic.java Hello World

⇒ It prints only Hello as we want arg[0] to print.

2. Prime Number.

```
import java.util.Scanner;  
  
public class Prime {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.println("Enter a number:");  
        int n = sc.nextInt();  
  
        if (n <= 1) {  
            System.out.println("Neither prime nor composite");  
        }  
  
        for(int c=2; c*c < n; c++){  
            if(n % c == 0){  
                System.out.println("Not a Prime");  
                return;  
            }  
        }  
        sc.close();  
        System.out.println("Prime");  
    }  
}
```

3. Hello World.

```
class Main {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

4. If Loop.

```
public static void If() {  
    int salary = 25400;  
    if (salary <= 10000) {  
        salary += 1000;  
    }  
    else if (salary <= 20000) {  
        salary += 2000;  
    }  
    else {  
        salary += 3000;  
    }  
    System.out.println(salary);  
}
```

5. While.

```
public static void While() {  
    Scanner sc = new Scanner(System.in);  
    System.out.print("Enter a number : ");  
    int n = sc.nextInt();  
    while (n <= 5) {  
        System.out.println(n);  
        n++;  
        sc.close();  
    }  
}
```

6. Do – While.

```
public static void DoWhile() {  
    Scanner sc = new Scanner(System.in);  
    System.out.print("Enter a number : ");  
    int n = sc.nextInt();  
    do {  
        System.out.println(n);  
        n++;  
    }  
    while (n <= 5);  
}
```

7. For.

```
public static void For() {  
    Scanner sc = new Scanner(System.in);  
    System.out.print("Enter a number : ");  
    int n = sc.nextInt();  
    for (int i = 0; i <= n; i++) {  
        System.out.println(i);  
    }  
    sc.close();  
}
```

8. Data Types.

```
public static void DT(){  
    byte b = 42;  
    char c = 'a';  
    short s = 1024;  
    int i = 50000;  
    float f = 5.67f;  
    double d = 0.1234;  
    double result = (f*b)+(i/c)-(d*s);  
    System.out.println((f*b)+" "+(i/c)+" "+" "+(d*s));  
    System.out.println(result);  
}
```

9. Sum.

```
static void sum(){
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter a Number : ");
    int a = sc.nextInt();
    System.out.print("Enter another Number : ");
    int b = sc.nextInt();
    int sum = a + b;
    System.out.println("The Sum is "+ sum);
}
```

10.Greet.

```
static void greet(){
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter your Name : ");
    String name = sc.nextLine();
    System.out.println("Hello "+ name);
}
```

11.Armstrong.

```
static boolean isArmstrong (int n) {
    int original = n;
    int sum = 0;

    while (n > 0){
        int rem = n % 10;
        n = n / 10;
        sum = sum + rem*rem*rem;
    }
    return sum == original;
}
```

12.Celcius to Farhenheit.

```
public static float ctof(){
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter temperature in Celsius: ");
    float c = sc.nextFloat();

    float f = (c * 9/5) + 32;
    System.out.print("Temperature in Fahrenheit: " + f);

    sc.close();
    return f;
}
```

13.Uppercase and Lowercase.

```
public static void ul() {
    Scanner sc = new Scanner (System.in);
    char ch = sc.next().trim().charAt(0);

    if (ch >= 'a' && ch <= 'z'){
        System.out.println("Lowercase");
    }
    else if (ch >= 'A' && ch <= 'Z') {
        System.out.println("Uppercase");
    }
    else {
        System.out.println("Not an alphabet");
    }
}
```

14.Fibonacci.

```
public static int fib() {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter the number of Fibonacci terms: ");
    int n = sc.nextInt();
    int a = 0;
    int b = 1;
    int count = 2;

    while (count <= n) {
        int temp = b;
        b = b+a;
        a = temp;
        count ++;
    }
    System.out.println("The " + n + "th Fibonacci number is: " + b);
}
```

15.Frequency.

```
public static int freq() {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter a number : ");
    int n = sc.nextInt();
    System.out.print("Enter the number to find frequency : ");
    int a = sc.nextInt();
    int count = 0;

    while (n > 0) {
        int rem = n % 10;
        if (rem == a) {
            count++;
        }
        n = n / 10;
    }
    System.out.println("Frequency of " + a + ": " + count);
    return count;
}
```

16.Reverse.

```
public static int rev() {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter a number : ");
    int n = sc.nextInt();
    int ans = 0;

    while (n > 0) {
        int rem = n % 10;
        n = n / 10;
        ans = ans * 10 + rem;
    }
    System.out.println("Reversed number: " + ans);
    return ans;
}
```

17. Calculator.

```
public static int calc() {
    Scanner sc = new Scanner(System.in);
    int ans = 0;
    while (true) {
        // take the operator as input
        System.out.print("Enter the operator: ");
        char op = sc.next().trim().charAt(0);

        if (op == '+' || op == '-' || op == '*' || op == '/' || op == '%')
        {
            // input two numbers
            System.out.print("Enter two numbers: ");
            int num1 = sc.nextInt();
            int num2 = sc.nextInt();

            if (op == '+') {
                ans = num1 + num2;
            }
            if (op == '-') {
                ans = num1 - num2;
            }
            if (op == '*') {
                ans = num1 * num2;
            }
            if (op == '/') {
                if (num2 != 0) {
                    ans = num1 / num2;
                }
            }
            if (op == '%') {
                ans = num1 % num2;
            }
        } else if (op == 'x' || op == 'X') {
            break;
        } else {
            System.out.println("Invalid operation!!");
        }
        System.out.println(ans);
    }
    return ans;
}
```

18.Switch Case.

```
public static int swtch() {  
    Scanner sc = new Scanner(System.in);  
    System.out.println("Enter a number : ");  
    int day = sc.nextInt();  
  
    switch (day) {  
        case 1 -> System.out.println("Monday");  
        case 2 -> System.out.println("Tuesday");  
        case 3 -> System.out.println("Wednesday");  
        case 4 -> System.out.println("Thursday");  
        case 5 -> System.out.println("Friday");  
        case 6 -> System.out.println("Saturday");  
        case 7 -> System.out.println("Sunday");  
        default -> System.out.println("Invalid day number");  
    }  
    return day;  
}
```